

# T-030 / R-039: the few-circles additive-energy bound is LINEAR in the

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

R-038 (dr2-t030-fewcircles) proved  $E_+(Q) \leq 3L^2N^2$  for  $Q$  covered by  $L$  circles, by Cauchy–Schwarz over the  $L^2$  index pairs plus the R-021 per-pair bound. The  $L^2$  factor is the obstruction to a sharper open-core statement. This note removes it: on a sphere the bound is *linear* in  $L$ . The operator direction was to sharpen the  $L$ -dependence so as to sharpen the cover-number lower bound on a hypothetical  $E_+ \geq N^{2+\delta}$  extremizer. Scope: arbitrary finite  $Q \subset S^2$ ; the H-NONLATTICE-REMAINDER frontier; NOT load-bearing for the published  $C_{\text{full}}$  head.

## 2. Theorem: the linear few-circles bound

**Theorem 1** (linear few-circles bound, sphere). *Let  $Q$  lie on an origin-centred sphere  $S^2$ ,  $N = |Q|$ , and be covered by  $L$  distinct circles. Then*

$$E_+(Q) \leq (2 + 2L)N^2 + 4LN + 4L^3 \leq 6(L + 1)N^2 \quad (L \leq N).$$

*Proof.* Write  $E_+ = r(0)^2 + \sum_{m \neq 0} r(m)^2$ ,  $r(m) = \#\{(a, b) \in Q^2 : a + b = m\}$ . For  $m \neq 0$  and a pair  $a + b = m$ , compute  $a \cdot m = a \cdot (a + b) = |a|^2 + a \cdot b$  and  $|m|^2/2 = \frac{1}{2}(|a|^2 + |b|^2) + a \cdot b$ ; hence  $a \cdot m = |m|^2/2 \iff |a| = |b|$ , which holds because all points of  $Q$  are at the same radius (origin-centred sphere). So both summands lie on the sum-level circle  $C_m = \{x \in S^2 : x \cdot m = |m|^2/2\}$  and  $r(m) \leq t_m := |Q \cap C_m|$ . Each cover circle  $\Gamma_i$  is a sphere section, hence  $\Gamma_i = C_{m_i}$  for a unique  $m_i \neq 0$  (a great circle is  $C_m$  for no  $m \neq 0$ ; its antipodal pairs feed only the  $m = 0$  term). Two distinct circles meet in  $\leq 2$  points, so for  $m \notin \{m_i\}$ ,  $t_m = \sum_i |Q_i \cap C_m| \leq 2L$ ; and  $t_{m_i} \leq \nu_i + 2L$  where  $\nu_i = |Q_i|$ . Therefore

$$\sum_{m \neq 0} r(m)^2 \leq 2L \sum_{m \neq 0} r(m) + \sum_i (\nu_i + 2L)^2 \leq 2LN^2 + (N^2 + 4LN + 4L^3),$$

using  $\sum_{m \neq 0} r(m) \leq N^2$ ,  $\sum_i \nu_i^2 \leq (\sum_i \nu_i)^2 = N^2$ ,  $\sum_i \nu_i = N$ , and  $|\{i\}| = L$ . Adding  $r(0)^2 \leq N^2$  gives the claim;  $4LN \leq 4N^2$  and  $4L^3 \leq 4LN^2$  for  $L \leq N$  yield  $6(L + 1)N^2$ .  $\square$

The exact bound improves R-038 for every  $L \geq 2$  (any  $N \geq 4$ ); the simplified  $6(L + 1)N^2$  form improves  $3L^2N^2$  for  $L \geq 3$  (e.g.  $L = 8$ :  $54N^2$  vs  $192N^2$ ; at  $L = 2$  via the exact  $6N^2 + 8N + 32 < 12N^2$ ). The per-pair R-021 input is replaced by the sharper sum-circle occupancy  $t_m \leq 2L$ , which is the multi-circle generalisation of Lemma A (R-025:  $E_+ \leq (1 + T')N^2$ , the single-richest-sum-circle case).

## 3. Sphere-essentiality (a recorded guard)

The identity  $a \cdot m = |m|^2/2 \iff |a| = |b|$  is the load-bearing step, and it requires all points equidistant from the centre. **Off the sphere the linear bound is not claimed (the proof mechanism  $r(m) \leq t_m$  fails):** a same-radius latitude *stack* (circles of equal radius at distinct heights – a cylinder,  $|a| \neq |b|$ ) has

product/AP structure with  $r(m) > t_m$  for vertical  $m$ , and indeed  $E_+ \sim 2LN^2$  there is consistent with linearity but the *proof* does not apply. The repository’s correct general-circle bound off the sphere is R-038’s quadratic  $3L^2N^2$  (whose R-021 input needs no equidistance). The proof-mechanism guard ( $\#\{m \neq 0 : r(m) > 2L\} \leq L$ , assert A6) FAILED on the first (cylinder) build and PASSES on the rebuilt genuine integer sphere  $x^2 + y^2 + z^2 = 1105^2$ ; the guard is retained as the falsifier of any off-sphere misuse.

#### 4. Consequence: a sharper cover-number lower bound and open core

**Corollary 1.** *If  $E_+(Q) \geq N^{2+\delta}$  for  $Q \subset S^2$ , then the circle-cover number of  $Q$  satisfies  $L \geq N^\delta/12$ .*

*Proof.*  $N^{2+\delta} \leq E_+ \leq 6(L+1)N^2 \leq 12LN^2$  ( $L \geq 1$ ).  $\square$

This doubles the exponent of R-038’s bound ( $L \geq N^{\delta/2}/\sqrt{3}$ ). Combined with R-033 Cor. 1.2 (an  $E_+ \geq N^{2+\delta}$  family carries  $\geq N^{2+\delta}/(8k^2 \log N)$  simultaneously  $k$ -rich *sum*-circles), the open core of T-030 is now pinned from two sides: a hypothetical extremizer must have BOTH a super-polynomial point-cover number ( $\geq N^\delta$  distinct cover circles) AND super-polynomially many rich sum-circles. The lacunary / few-circle families (R-038, technique-only witness) are excluded; the residual is squarely the super-polynomially-many-circles regime, an open circle-incidence problem.

#### 5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- ( $\alpha$ ) “*Just a multi-circle restatement of Lemma A.*” VALID: the occupancy bound  $t_m \leq 2L$  off  $\leq L$  sum-circles is exactly the multi-circle generalisation of Lemma A’s single- $T'$  bound. The new content is the explicit  $L$ -linear closed form and the doubled cover-number lower bound (Cor.), strictly improving R-038.
- ( $\beta$ ) “*Is the linear  $L$ -factor tight on  $S^2$ ?*” UPHELD as open, and NOT claimed: on the genuine sphere the measured energy of rich-latitude unions is  $E_+/N^2 \in \{2.97, 2.59, 2.42, 2.32, 2.27, 2.22\}$  for  $L = 1 \dots 8$  – essentially  $L$ -INDEPENDENT ( $\sim 3N^2$ ), far below  $6(L+1)N^2$ . The curvature suppresses the energy (consistent with R-033’s global  $N^{9/4}$  ceiling); the linear bound is correct but loose on these families, and its  $L$ -dependence may not be saturable on  $S^2$ . No tightness is claimed.
- ( $\gamma$ ) “*Sphere-essential – so the cylinder evidence is spurious.*” VALID and documented (Sec. 3): the first test build was a cylinder, the guard A6 caught the off-sphere  $r(m) > t_m$ , and the configs were rebuilt on a genuine sphere where A6 passes. Off the sphere R-038’s quadratic stands; the linear bound is asserted only for  $Q \subset S^2$ , which is the T-030 setting.
- ( $\delta$ ) “*Great circles break the  $m_i$  enumeration.*” DISMISSED: a great circle is  $C_m$  for no  $m \neq 0$ , so it contributes  $\leq 2$  to every  $t_m$  ( $m \neq 0$ ) and feeds only the split-off  $r(0)^2 \leq N^2$  term; the  $\leq L$  special sum-circles are exactly the non-great cover circles.

**Quantitative sanity check (mandatory).** *Mechanism.* On the genuine sphere,  $\#\{m \neq 0 : r(m) > 2L\} \leq L$  for  $L = 2, 3, 4, 6, 8$  (assert A6) – the proof’s occupancy split holds. *Magnitude.*  $L = 1$ :  $E_+/N^2 = 2.97$  (the rich latitude,  $N = 108$ ,  $E_+ = 34668$ , matching R-033 F1)  $\leq 6 \cdot 2 = 12$ . *Limit / improvement.*  $B_{\text{lin}} < 3L^2N^2$  for all  $L \geq 2$  (assert A2). *Estimator.* vectorised  $E_+ =$  brute  $O(N^4)$  (396, assert A8); all  $L = 1$  points verified on  $x^2 + y^2 + z^2 = 1105^2$ . *Reproducibility.* deterministic exact-integer; exits 0 iff all 6 asserts pass.

## 6. Result footer

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Result ID:              | R-039 (linear few-circles bound, sphere) / T-030 frontier                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Precise statement:      | Q on an origin-centred $S^2$ , covered by L distinct circles $\Rightarrow E_+(Q) \leq (2+2L)N^2 + 4LN + 4L^3 \leq 6(L+1)N^2$ . LINEAR in L; strictly improves R-038's $3L^2N^2$ for $L \geq 2$ . Proof: sum-circle occupancy $t_m \leq 2L$ off the $\leq L$ cover sum-circles (multi-circle Lemma A / R-025 refinement); SPHERE-ESSENTIAL ( $r(m) \leq t_m$ needs $ a = b $ ). Cor: $E_+ \geq N^{2+\delta} \Rightarrow$ cover number $L \geq N^\delta/12$ (vs R-038's $N^{\delta/2}/\sqrt{3}$ ; exponent doubled). |
| Scope:                  | T-030 (arbitrary finite Q in $S^2$ ; H-NONLATTICE-REMAINDER). NOT load-bearing for the $C_{\text{full}}$ head. Off the sphere the linear bound is NOT CLAIMED (proof mechanism fails); R-038's quadratic is the guaranteed general-circle bound there.                                                                                                                                                                                                                                                             |
| Evidence grade:         | PARTIAL ADVANCE. Theorem T7-WITHIN-CLASS (elementary, sphere). Strictly improves R-038. Tightness on $S^2$ NOT claimed (measured $E_+ \sim 3N^2$ , L-independent on rich-latitude unions). Does NOT close T-030: arbitrary-Q $N^{2+\epsilon}$ STILL OPEN (record $N^{9/4}$ , R-033).                                                                                                                                                                                                                               |
| Reproduction command:   | python3 codes/vacuum/dr2_t030_fewcircles_linear.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Expected output:        | 6/6 asserts PASS, exit 0; JSON at claims/B5-BEYOND-LAYER-BOUND/runs/260614-dr2-t030-fewcircles-linear/result.json                                                                                                                                                                                                                                                                                                                                                                                                  |
| Falsifier:              | (a) Q on L distinct circles of a sphere with $E_+ > 6(L+1)N^2$ ; (b) $\#\{m \neq 0 : r(m) > 2L\} > L$ on a genuine sphere config (would break the occupancy split). Both machine-checked false on $S^2$ ; off the sphere the proof mechanism $r(m) \leq t_m$ fails, but the cylinder STILL SATISFIES the bound (no counterexample); thus scoped to $S^2$ .                                                                                                                                                         |
| Tier before / after:    | NONE -- operator decides. No gate/tier/hypothesis flip. B5 stays T7-SCOPE_{admissibility-bounded} given A1; T-030 OPEN; DR2-SHARE RESCOPED-T0-T030-NONLOADBEARING.                                                                                                                                                                                                                                                                                                                                                 |
| No-overclaim statement: | Strict improvement of R-038's bound, sphere-only; the L-linear factor is NOT shown tight on $S^2$ (likely loose; true energy may be $O(N^2 \text{ polylog})$ per R-033). Does NOT close T-030. Builds on Lemma A (R-025) and R-038.                                                                                                                                                                                                                                                                                |
| Next required action:   | Operator review. Continuation: (i) determine the true $S^2$ L-dependence (is $E_+ = O(N^2 \text{ polylog})$ for any bounded-curvature-circle union?); (ii) attack the super-poly-circle core via improved circle incidences (R-033 route).                                                                                                                                                                                                                                                                         |